

What is discrete mathematics?

Discrete mathematics is a branch of mathematics that deals with **mathematical structures** and objects that are fundamentally discrete or separate in nature.

In mathematics, a structure is a set endowed (সমৃদ্ধ) with some additional features on the set (e.g. an operation, relation, metric, or topology). Often, the additional features are attached or related to the set, so as to provide it with some additional meaning or significance.

What is discrete mathematics?

Discrete mathematics is the part of mathematics devoted to the study of discrete objects. Discrete mathematics is the branch of mathematics dealing with objects that can assume only distinct, separated values. The term "discrete mathematics" is therefore used in contrast with "continuous mathematics," which is the branch of mathematics dealing with objects that can vary smoothly (which includes calculus). Whereas discrete objects can often be characterized by integers, continuous objects require real numbers.

Discrete mathematics or finite mathematics is the branch of mathematics in which the mathematical organizations are fundamentally isolated, that is, the notion of continuity does not apply to them. The formulas of continuity do not apply to it. This is why the name has become Discrete mathematics. Most or all of the objects studied in Discrete mathematics are computational sets, such as integers, finite graphs, and statistical languages.

Discrete mathematics has wide-ranging applications in computer science, cryptography, information theory, operations research, optimization, and many other fields. It provides the foundation for solving real-world problems using rigorous (কঠোর) mathematical techniques and logical reasoning.

Why we study Discrete Mathematics

We study **Discrete Mathematics** because it builds the foundation for computer science, cryptography, and algorithms, enabling problem-solving with finite, countable structures. It develops logical thinking and precision, essential for programming and data analysis. Many modern technologies from internet security to AI all rely directly on its concepts.

Why we study Discrete Mathematics

Foundation for Computer Science: Discrete mathematics forms the backbone of computer science. It provides the fundamental mathematical tools and concepts used in algorithm design, data structures, logic, cryptography, and other areas of computer science. Understanding discrete mathematics is crucial for developing efficient algorithms, analyzing their complexity, and solving computational problems.



Problem Solving and Logical Reasoning: Discrete mathematics enhances problem-solving skills and logical reasoning abilities. It emphasizes rigorous (কঠোর) thinking, proof techniques, and logical deductions. Studying discrete mathematics trains individuals to think critically, analyze problems, and construct logical arguments—an essential skill set for a wide range of disciplines.

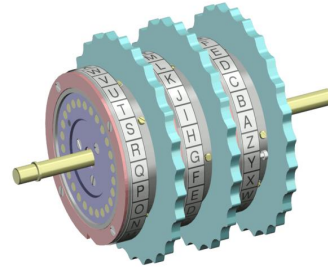


Cryptography and Information Security: Discrete mathematics is central to cryptography, the science of secure communication. Concepts such as modular arithmetic, number theory, and Boolean algebra are essential in designing and analyzing cryptographic algorithms. Discrete mathematics also helps in understanding and developing protocols for secure data transmission, authentication, and encryption.



Data Analysis and Network Theory
Decision-Making and Operations Research

Overall, studying discrete mathematics helps in developing analytical thinking, problem-solving skills, and mathematical maturity. Its applications extend to computer science, cryptography, data analysis, network theory, optimization, and other disciplines, making it an essential subject for students pursuing careers in STEM fields and related areas.



STEM (Science, Technology, Engineering and Mathematics) career

In computer science, cryptography refers to secure information and communication techniques derived from mathematical concepts and a set of rule-based calculations called algorithms, to transform messages in ways that are hard to decipher.

<https://cryptii.com/>

Cryptography is the science of protecting information by transforming it into a secure format so that only authorized parties can read or understand it.

It involves two main processes

Encryption: Converting plain text into unreadable code (cipher text)

Decryption: Converting cipher text back into the original plain text

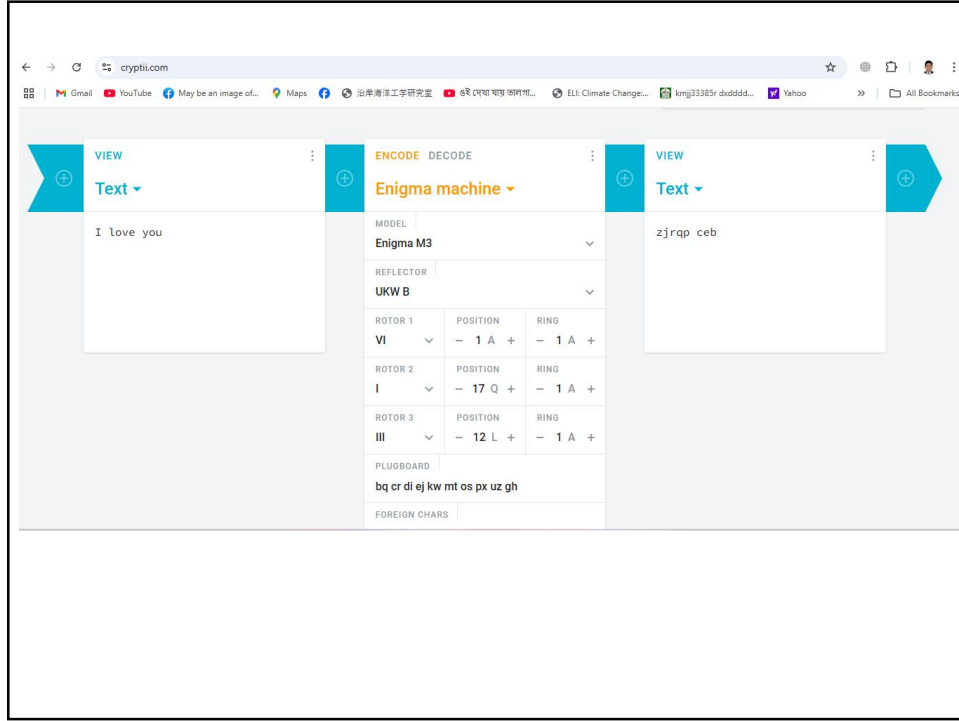
Number Theory: Prime numbers and modular arithmetic are used in RSA, Diffie-Hellman, and similar systems.

Logic: Boolean operations are used in encryption algorithms.

Combinatorics: Used to calculate the number of possible keys and determine password complexity.

Graph Theory: Applied in network security and data routing.

Probability and Statistics: Required for random number generation and security analysis.



- ১। ক্রিপ্টোগ্রাফি মানে হলো গোপন বার্তা আদান-প্রদানের পদ্ধতি।
- ২। প্রাচীন মিশরে গোপন হারারোগ্লিফ ব্যবহার হতো।
- ৩। গ্রীকরা স্কিটেলি (Scytale) নামক যন্ত্র দিয়ে বার্তা লুকাতো।
- ৪। জুলিয়াস সিজার "সিজার সাইফার" ব্যবহার করতেন, যেখানে অক্ষর স্থানান্তরিত হতো।
- ৫। ৯ম শতকে আল-কিন্দি ফ্রিকোয়েন্সি অ্যানালাইসিস পদ্ধতি উদ্ভাবন করেন।
- ৬। ১৬শ শতকে ভিজেনোর সাইফার তৈরি হয়, যা আরও জটিল।
- ৭। দ্বিতীয় বিশ্বযুদ্ধে জার্মানির এনিগমা মেশিন ভেঙে ফেলেন অ্যালান ট্যুরিং।
- ৮। ১৯৭৬ সালে পাবলিক কি ক্রিপ্টোগ্রাফি আবিষ্কৃত হয়।
- ৯। আধুনিক যুগে RSA, AES, এবং ECC খুবই জনপ্রিয় এনক্রিপশন পদ্ধতি।
- ১০। বর্তমানে ক্রিপ্টোগ্রাফি ইন্টারনেট, ব্যাংকিং ও সাইবার নিরাপত্তায় গুরুত্বপূর্ণ ভূমিকা রাখছে।

What is a proposition?

An assertion is a statement. A proposition is an assertion which is either true or false (but not both).

The Following are propositions:

- ▶ 4 is a prime number
- ▶ $3 + 3 = 6$
- ▶ The moon is made of cheese.

The sky is blue. (True)

$2 + 2 = 5$. (False)

Dhaka is the capital of Bangladesh. (True)

Water boils at 50°C . (False)

The following are not propositions:

- $x+y > 4$. Is this true or false? It depends on the value of x and y , the statement takes a true or false value.
- $x=3$. You cannot associate a truth value to this because it simply assigns a value to x .
- Are you leaving? This is not an assertion, it is a question.
- Buy 4 Books this is not an assertion, it is an order.

Types of Propositions

1.Atomic Proposition: A simple proposition that cannot be broken down further.

1.Example: "It is raining."

2.Compound Proposition: Formed by combining simple propositions using logical connectives ($\neg$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$).

1.Example: "It is raining **and** it is cold."

The negation

The **negation** of statement p is "not p ." The negation of p is symbolized by " $\sim p$ or $\neg p$." The truth value of $\sim p$ is the opposite of the truth value of p .

p :	The number 9 is odd.	true
$\sim p$:	The number 9 is not odd.	false

Construct a truth table for the negation of p .

Solution:

p	$\sim p$
T	F
F	T

Given:	a: A triangle is not a polygon.
	b: A square is a rectangle.
Problem:	Which of the following is the negation of "A triangle is not a polygon"?

Disjunction

A disjunction is a compound statement formed by joining two statements with the connector OR. The disjunction "p or q" is symbolized by $p \vee q$. **A disjunction is false if and only if both statements are false; otherwise it is true.** The truth values of $p \vee q$ are listed in the truth table below.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

p: "7 is greater than 5." (True)

q: "10 is less than 3." (False)

$p \vee q$: "7 is greater than 5 **or** 10 is less than 3."

• Since p is True, the whole disjunction is True.

p: "এই জামাটি লাল রঙের।" (সত্য)

q: "এই জামাটি নীল রঙের।" (মিথ্যা)

$p \vee q$: "এই জামাটি লাল **অথবা** নীল।"

এর মানে — যতক্ষণই জামাটি লাল, বাকিটা যতই মিথ্যা হোক না কেন, বাক্যটি সত্য।

or

Let p and q be propositions. The disjunction of p and q , denoted by $p \vee q$, is the proposition “ p or q .” The disjunction $p \vee q$ is false when both p and q are false and is true otherwise.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Example Complete a truth table for
: each disjunction below.

1. a or b
2. a or not b
3. not a or b

a	b	$a \vee b$	a	b	$\sim b$	$a \vee \sim b$	a	b	$\sim a$	$\sim a \vee b$
T	T	T	T	T	F	T	T	T	F	T
T	F	T	T	F	T	T	T	F	F	F
F	T	T	F	T	F	F	F	T	T	T
F	F	F	F	F	T	T	F	F	T	T

Conjunction

A **conjunction** is a compound statement formed by joining two statements with the connector AND. The conjunction "p and q" is symbolized by $p \wedge q$. A conjunction is true when both of its combined parts are true; otherwise it is false.

we complete the last column according to the rules for conjunction listed above.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

or

Let p and q be propositions. The conjunction of p and q, denoted by $p \wedge q$, is the proposition "p and q." The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p: "It is sunny." (True)

q: "I am wearing sunglasses." (True)

Conjunction: $p \wedge q$: "It is sunny **and** I am wearing sunglasses."

Both are true, whole statement is **True**

p: "আজ রবিবার।" (সত্য)

q: "আমি আজ ক্লাসে যাব।" (সত্য)

$p \wedge q$: "আজ রবিবার এবং আমি আজ ক্লাসে যাব।"

•যেহেতু দুটোই সত্য, তাই পুরো বক্তব্য সত্য।

Example : Construct a truth table for each conjunction below:

1. x and y

2. $\sim x$ and y

Solution: 3. $\sim y$ and x

x	y	$x \wedge y$	x	y	$\sim x$	$\sim x \wedge y$	x	y	$\sim y$	$\sim y \wedge x$
T	T	T	T	T	F	F	T	T	F	F
T	F	F	T	F	F	F	T	F	T	T
F	T	F	F	T	T	T	F	T	F	F
F	F	F	F	F	T	F	F	F	T	F

Exclusive Or

Let p and q be propositions. The exclusive or of p and q , denoted by $p \oplus q$, is the proposition that is true when exactly one of p and q is true and is false otherwise.

P	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

p : “ $2 + 3 = 5$ ” (True)

q : “ $4 \times 2 = 9$ ” (False)

$p \oplus q$: “ $2+3=5$ XOR $4 \times 2=9$ ”

Exactly one is true $p \oplus q$ is **True**

p : “আমি চা খাই।”

q : “আমি কফি খাই।”

$p \oplus q$: “আমি চা বা কফি খাই, কিন্তু উভয় একসাথে নয়।”

যদি আপনি শুধু চা খান $\rightarrow$ সত্য

যদি শুধু কফি খান $\rightarrow$ সত্য

যদি দুটোই খান $\rightarrow$ মিথ্যা

যদি দুটোই না খান $\rightarrow$ মিথ্যা

Conditional Statements

Let p and q be propositions. The conditional statement $p \rightarrow q$ is the proposition “if p , then q .” The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise. In the conditional statement $p \rightarrow q$, p is called the hypothesis and q is called the conclusion.

The statement $p \rightarrow q$ is called a conditional statement because $p \rightarrow q$ asserts that q is true on the condition that p holds. A conditional statement is also called an implication.

P	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Bi-conditional

Let p and q be propositions. The bi-conditional statement $p \leftrightarrow q$ is the proposition “ p if and only if q .” The bi-conditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise. Bi-conditional statements are also called bi-implications.

P	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Tautology.

A compound statement, that is always true regardless of the truth value of the individual statements, is defined to be a **tautology**.

Let's look at another example of a tautology.

Example 2: Is $(p \wedge q) \rightarrow p$ a tautology?

p	q	$p \wedge q$	$(p \wedge q) \rightarrow p$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

Example 3: Is $x \rightarrow (x \vee y)$ a tautology?

x	y	$x \vee y$	$x \rightarrow (x \vee y)$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

Solution: Yes; the truth values of $x \rightarrow (x \vee y)$ are {T, T, T, T}.

Example 5: Is $(p \vee q) \rightarrow (p \wedge q)$ a tautology?

p	q	$(p \vee q)$	$(p \wedge q)$	$(p \vee q) \rightarrow (p \wedge q)$
T	T	T	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	F	T

Solution: No; the truth values of $(p \vee q) \rightarrow (p \wedge q)$ are {T, F, F, T}.

Example 6: Is $[(p \rightarrow q) \wedge p] \rightarrow p$ a tautology?

p	q	$p \rightarrow q$	$(p \rightarrow q) \wedge p$	$[(p \rightarrow q) \wedge p] \rightarrow p$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

Solution: Yes; the truth values of $[(p \rightarrow q) \wedge p] \rightarrow p$ are {T, T, T, T}.

Example 7: Is $(r \rightarrow s) \leftrightarrow (s \rightarrow r)$ a tautology?

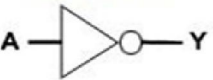
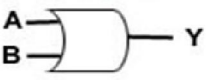
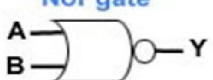
r	s	$r \rightarrow s$	$s \rightarrow r$	$(r \rightarrow s) \leftrightarrow (s \rightarrow r)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

Solution: No; the truth values of $(r \rightarrow s) \leftrightarrow (s \rightarrow r)$ are {T, F, F, T}.

Contradiction: A proposition that is false under all circumstances is called Contradiction. The truth table

p	$\sim p$	$p \wedge \sim p$
T	F	F
F	T	F

Logic Gate Examples

Not gate		
	If A = 0 then Y = 1 If A = 1 then Y = 0	Negation
Or gate		
	If A = 0 and B = 0 then Y = 0 If A = 1 and B = 0 then Y = 1 If A = 0 and B = 1 then Y = 1 If A = 1 and B = 1 then Y = 1	$p \vee q$ Disjunction
And gate		
	If A = 0 and B = 0 then Y = 0 If A = 0 and B = 1 then Y = 0 If A = 1 and B = 0 then Y = 0 If A = 1 and B = 1 then Y = 1	$p \wedge q$ Conjunction
Nand gate		
	If A = 0 and B = 0 then Y = 1 If A = 0 and B = 1 then Y = 1 If A = 1 and B = 0 then Y = 1 If A = 1 and B = 1 then Y = 0	$\sim(p \wedge q)$
Nor gate		
	If A = 0 and B = 0 then Y = 1 If A = 0 and B = 1 then Y = 0 If A = 1 and B = 0 then Y = 0 If A = 1 and B = 1 then Y = 0	$\sim(p \vee q)$

